4-9-26 (WEEK 7)

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Continuing from the example last time,

$$f_X(t) = \begin{cases} \frac{2}{t^3}, & t \geq 1 \\ 0, & t < 1 \end{cases}$$

Then,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,dt = \int_1^{\infty} t \cdot \frac{2}{t^3}\,dt = \left[-\frac{2}{t}\right]_1^{\infty} = 2$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \int_{-\infty}^{\infty} t^2 f_X(t)\,dt - 2^2 = \int_1^{\infty} t^2 \cdot \frac{2}{t^3}\,dt - 4 = \left[2\ln|t|\right]_1^{\infty} - 4 = \infty$$

$\Rightarrow$ Variance DNE

Note: The divergence of variance can even occur for discrete random variables if the probability of large values doesn't decay quickly

E.g. $\mathbb{P}(X=k) = \frac{c}{k^2}$, $c = \frac{6}{\pi^2}$. Then, $\mathbb{E}[k^2] = \infty$

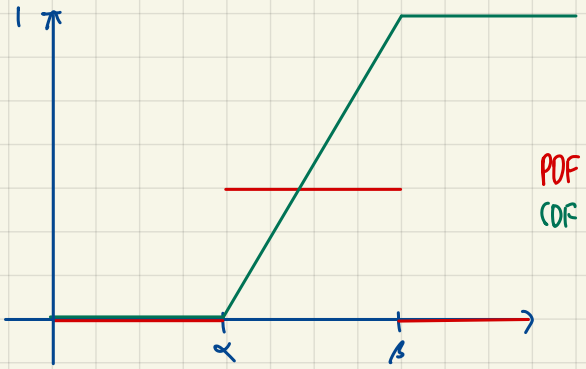
# CONTINUOUS DISTRIBUTIONS

## UNIFORM DISTRIBUTION

Say $\alpha < \beta \in \mathbb{R}$, $X \sim \mathrm{Unif}(\alpha, \beta)$

Setting: Want a uniformly random point in $[\alpha, \beta]$

$\Rightarrow$ Probability of a subinterval depends on its width




PDF (Prob. Density Func)

$$f_X(t) = \begin{cases} c, & t \in [\alpha, \beta] \\ 0, & t \notin [\alpha, \beta] \end{cases}$$

Then, $1 = \int_{-\infty}^{\infty} f_X(t)\,dt = \int_\alpha^\beta c\,dt = c(\beta - \alpha)$

$\therefore$ $c = \frac{1}{\beta - \alpha}$

CDF (Cumulative Density Func)

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$

$$= \begin{cases} 0, & x \leq \alpha \\ \int_\alpha^x \frac{1}{\beta - \alpha}\,dt = \frac{x - \alpha}{\beta - \alpha}, & \alpha < x \leq \beta \\ 1, & x > \beta \end{cases}$$

Expectation

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,dt = \int_\alpha^\beta \frac{t}{\beta - \alpha}\,dt = \left[\frac{t^2}{2(\beta - \alpha)}\right]_\alpha^\beta = \frac{\alpha + \beta}{2}$$

Variance

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \int_{-\infty}^{\infty} t^2 f_X(t)\,dt - \left(\frac{\alpha+\beta}{2}\right)^2 = \int_\alpha^\beta \frac{t^2}{\beta - \alpha}\,dt - \left(\frac{\alpha+\beta}{2}\right)^2 = \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \frac{\alpha^2 + 2\alpha\beta + \beta^2}{4} = \frac{(\beta - \alpha)^2}{12}$$

## THEOREM (SIMULATING CONTINUOUS RANDOM VARIABLES)

Let $X$ be a continuous random variable with cdf $F_X : \mathbb{R} \to [0,1]$. Let $U \sim \mathrm{Unif}(0,1)$, $Y = F_X^{-1}(U) \in \mathbb{R}$. Then, $Y$ has the same distribution as $X$

Proof

We want to show "$F_Y = F_X$". Then, $F_Y(x) = \mathbb{P}(Y \leq x) \overset{\text{def of } Y}{=} \mathbb{P}(F_X^{-1}(U) \leq x) \overset{F_X \text{ increasing}}{=} \mathbb{P}(U \leq F_X(x)) \overset{U \sim \mathrm{Unif}(0,1)}{=} F_U(F_X(x)) = \frac{F_X(x) - 0}{1 - 0} = F_X(x)$ $\square$

# EXPONENTIAL DISTRIBUTION

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no 67 :)

Setting: A bus arrives at a bus stop on average every 18 mins. We arrive at the bus stop. How long do we have to wait for the next bus?

given we waited x min, prob (we wait x+y min) = prob (somebody just arrives and needs to wait for y mins)

If $X$ is our waiting time, we want $X$ to satisfy $\mathbb{P}(X \geq x+y \mid X \geq x) = \mathbb{P}(X \geq y)$

This is called the "memoryless property"

## THEOREM

The only continuous memoryless distribution is the exponential distribution.

Proof

Let $X$ be a memoryless continuous distribution, define $g(x) := \mathbb{P}(X \geq x)$

Then, $g(x+y) = \mathbb{P}(X \geq x+y)$

(def of cond. prob.) $= \mathbb{P}(X \geq x+y \mid X \geq x)\mathbb{P}(X \geq x)$

(assumption) $= \mathbb{P}(X \geq y)\mathbb{P}(X \geq x)$

$= g(x)g(y)$

$\therefore g(x+y) = g(x)g(y) \quad \forall x, y \geq 0$

Define $\lambda := -\log g(1) \Leftrightarrow g(1) = e^{-\lambda}$

Notice, $g(1) = \mathbb{P}(X \geq 1) \in [0,1] \Rightarrow \lambda \in (0, \infty)$

Let $n \in \mathbb{N}$, then $g(n) = g(\overbrace{1 + \dots + 1}^{n \text{ times}}) = g(1)^n = \boxed{e^{-n\lambda}}$

Let $\frac{r}{s} \in \mathbb{Q}_{\geq 0}$, then $g(r) = g(\overbrace{\tfrac{r}{s} + \dots + \tfrac{r}{s}}^{s \text{ times}}) = g(\tfrac{r}{s})^s \Rightarrow g(\tfrac{r}{s}) = g(r)^{\frac{1}{s}} = \boxed{e^{-\frac{r}{s}\lambda}}$

Let $x \in \mathbb{R}$. Let $q_n$ be a seq of rational values with $q_n \searrow x$.

Then, $g(x) = \mathbb{P}(X \geq x) = 1 - \mathbb{P}(X \leq x)$ right-cont. $\because$ CDF

$= 1 - \lim_{n \to \infty} \mathbb{P}(X \leq q_n)$

$= \lim_{n \to \infty} g(q_n)$

$= \lim_{n \to \infty} e^{-\lambda q_n}$

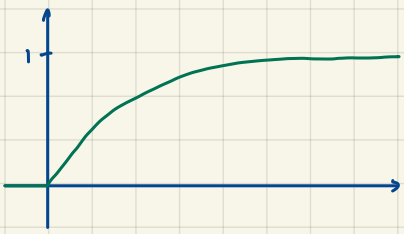
$= e^{-\lambda x}$

## DEFINITION (EXPONENTIAL DISTRIBUTION)

We denote $X \sim \mathrm{Exp}(\lambda)$, $\lambda > 0 \in \mathbb{R}$

CDF:

$$\mathbb{P}(X \leq x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$




PDF:

$$f_X(t) = F_X'(t) = \begin{cases} \lambda e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Expectation: $\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t)\,dt = \int_0^{\infty} t\lambda e^{-\lambda t}\,dt = \cancel{[t(-e^{-\lambda t})]_0^{\infty}}^{\,0} - \int_0^{\infty} (-e^{-\lambda t})\,dt = \left[-\frac{e^{-\lambda t}}{\lambda}\right]_0^{\infty} = \frac{1}{\lambda}$ (Poisson is $\frac{1}{\lambda}$)

Variance: $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \int_{-\infty}^{\infty} t^2 f_X(t)\,dt - \frac{1}{\lambda^2} = \int_0^{\infty} t^2 \lambda e^{-\lambda t}\,dt - \frac{1}{\lambda^2} = \cancel{[t^2(-e^{-\lambda t})]_0^{\infty}}^{\,0} - \int_0^{\infty} 2t(-e^{-\lambda t})\,dt - \frac{1}{\lambda^2} = 2\int_0^{\infty} te^{-\lambda t}\,dt - \frac{1}{\lambda^2} = \frac{2}{\lambda}\int_0^{\infty} te^{-\lambda t}\,dt - \frac{1}{\lambda^2} = \frac{2}{\lambda}\left(\frac{1}{\lambda}\right) - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$

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## EXAMPLE

A bus arrives every 15 mins. What is the probability that we have to wait $\geq 5$ mins for the bus?

Solution

$X \sim Exp(\lambda)$

We know $\mathbb{E}(X) = \frac{1}{\lambda} = 15 \Rightarrow \lambda = \frac{1}{15}$

$\therefore \mathbb{P}(X \geq 5) = e^{-\lambda(5)} = e^{-\frac{1}{3}}$

## EXAMPLE

A passenger can take one of two buses.

The first bus arrives every 15 min on avg; the second bus arrives every 18 min on avg.

If the passanger takes the first bus that arrives, and they run indep, how long will they have to wait on avg?

Solution

Let $X_1$ be the waiting time for the first bus $\Rightarrow X_1 \sim Exp(\frac{1}{15})$

Let $X_2$ be the waiting time for the second bus $\Rightarrow X_2 \sim Exp(\frac{1}{18})$

Let $W$ = waiting time for the passenger (the first bus to show up)

We see $W = \min\{X_1, X_2\}$. Consider $x \geq 0$.

Then, $F_W(x) = \mathbb{P}(W \leq x) = \mathbb{P}(\min\{X_1, X_2\} \leq x) = 1 - \mathbb{P}(\min\{X_1, X_2\} \geq x) = 1 - \mathbb{P}(X_1 \geq x \cap X_2 \geq x)$

$= 1 - \mathbb{P}(X_1 \geq x)\mathbb{P}(X_2 \geq x)$ (by indep)

$= 1 - e^{-\frac{1}{15}x} e^{-\frac{1}{18}x}$

$= 1 - e^{-(\lambda_1 + \lambda_2)x}$, $\lambda_1 = \frac{1}{15}$, $\lambda_2 = \frac{1}{18}$, another exp. dst!!

$\therefore W \sim Exp(\lambda_1 + \lambda_2) = Exp(\frac{11}{90})$

$\therefore \mathbb{E}(W) = \frac{90}{11}$

More generally, if $X_i \sim Exp(\lambda_i)$, $1 \leq i \leq n$, each are indep, then $X = \min\{X_i\} \sim Exp(\sum_{i=1}^{n} \lambda_i)$